Herald

Repository: `git@github.com:vasic-digital/Herald.git`

Purpose

Ingesting system events and reliably fanning them out to multiple notification channels so every alert reaches the right destination without confusion.

What Herald can do?

Herald is the mechanism which receives input from the source and sends it to one or more destinations. Depending on the implementation (flavor) of the Herald sources and destinations could be various. We can have single type or multiple type inputs and same for the outputs.

For example input can be the result of execution of the pipeline and the output sending of this result (some report for example) to certain output - messagin system which notifies subscribers!

Possibilities are not limited.

The structure of the System MUST be hierarchical so in the top levels (closest to the root) we have abstractions and base reusable, main shared mechanisms, while inside the `flvors` directory we MUST have all flavors - the implementations.

Note: We MUST NOT be obligated to follow this structure as many parent project specific flavors (which may be propriatery) may exist! We MUST make sure that such flexibility is possible!

How Herald should achieve its mission objectives?

Every herald flavor will be individual binnary which users can add to the System path throguh the `.bashrc` or `.zshrc` (for example). Each Herald flavor will have shared set of commands and parameters while there will be as well commands and parameters specific to particular Herald flavors.

Herald applications are CLI binarries which are mainly designed for CI integration and various Pipelines. They can be easily incorporate for use with various AI CLI Agents as well or any similar use cases (triggering by Crond and so on ...).

Some Herald application names would be: `pherald` for the `Project Herald`, `sherald` for the `System Herald` and so on.

Technology stack

Herald project and all flavors MUST BE writtn in Go.

Flavors (the implementations)

Main Go abstractions and shared codebase (with shared implementations) will be used as the base which Flavors will inherit and build on top of it.

Project Herald

The Project Herald is focused on Projects and its development. All Projects share some commons and Project Herald MUST FIT as the universal player here! Project Herald MUST offer several integrations:

- Telegram
- Slack
- Max (max.ru)
- Email
- Markdown document with PDF and HTML export

For each of Messaging services user MUST provide required tokens, credentials or API keys depending on the platform. All details MUST BE documented in proper user guides and manuals with step by step instructions so users can easily obtain and provide required information.

All sensitive data like credentials, tokens or API keys MUST be in inside proper .env file (create for the documentation purposes .env.example file to illustrate everything that users can put there) which MUST BE Git ignored! System MUST BE capable to obtain all these environment variables from exported variables from `.bashrc` and `.zshrc` or any other System profile script which can export the variables. Using values fron .env comes after top level ones are loaded and everything defined inside the .env file will override them (if same variabels are defined there).

Everything that is sent or received through any of the interated Messnger channels such as Telegram, Slack, Max and others (Email as well) will be stored inside main Markdown file and exported into PDF and HTML regularly. Markdown file and its exports MUST BE always in sync! Location of the Markdown file inside the Project MUST BE the following: `docs/herald/diary/main.md` (`main.pdf` and `main.html`).

System Herald

Tbd

Others and misc

Here we will list main ideas for upcoming Flavors which MUST BE planned with proper deep web research and fully implemented:

- Tbd

Integration into the Constitution

Tbd

Notes

Tbd